



# End-to-End User Journeys & Operational Flows

20 Detailed Business Workflows, Preconditions & Step-by-Step Sequences

| DOCUMENT TYPE         | TARGET AUDIENCE                        | RELEASE VERSION   | STATUS                |
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## Complete User Flows

This document details every major user flow step-by-step, including preconditions, postconditions, expected results, and related APIs.

### 1. User Registration & Onboarding Flow

- **Preconditions:** User has a valid phone number.
- **Actor:** New User/Shop Owner
- **Steps:**
  1. Enter name, phone, email, shop name.
  2. Request OTP.
  3. Enter OTP and set password.
- **Expected Results:** Shop created, User created as Owner, JWT token returned.
- **Error Scenarios:** Invalid OTP, Phone already exists.
- **APIs:** POST /api/auth/register , POST /api/auth/verify-otp

### 2. Login & Authentication Flow

- **Preconditions:** Account exists and is verified.
- **Actor:** Any User
- **Steps:**
  1. Enter phone and password.
  2. Click login.
- **Expected Results:** JWT access and refresh tokens returned, User profile loaded.
- **Error Scenarios:** Invalid credentials, Inactive account.
- **APIs:** POST /api/auth/token

### 3. POS Checkout Flow

- **Preconditions:** Cash drawer open, products in stock.

- **Actor:** Cashier
- **Steps:**
  1. Scan barcode or search product.
  2. Add to cart, adjust qty/price.
  3. Select/add customer.
  4. Select payment method(s) and enter amounts.
  5. Submit sale.
- **Expected Results:** Sale created, stock deducted, ledger updated, invoice printed.
- **Error Scenarios:** Insufficient stock, Invalid payment amount.
- **APIs:** `GET /api/pos/products` , `POST /api/pos/checkout`

#### 4. Product Creation Flow

- **Preconditions:** User has product management permission.
- **Actor:** Manager/Admin
- **Steps:**
  1. Navigate to Products > Add New.
  2. Enter name, category, brand, cost, price.
  3. Toggle `is_serialized` or `track_inventory` .
  4. Save.
- **Expected Results:** Product record created.
- **Error Scenarios:** Duplicate SKU.
- **APIs:** `POST /api/inventory/products`

#### 5. Product Purchase/Stock-In Flow

- **Preconditions:** Supplier exists.
- **Actor:** Manager/Admin
- **Steps:**
  1. Select Supplier.
  2. Add products, define cost and qty.
  3. If serialized, enter/scan serial numbers.
  4. Enter payment amount (full, partial, or credit).
  5. Save Purchase.
- **Expected Results:** Stock increased, Supplier payable updated, Ledger updated.
- **Error Scenarios:** Missing serials for serialized item.
- **APIs:** `POST /api/inventory/purchases`

#### 6. Sales Return & Refund Flow

- **Preconditions:** Original sale exists.
- **Actor:** Cashier/Manager

- **Steps:**
  1. Lookup invoice.
  2. Select items to return and condition (restock or damage).
  3. Process refund amount.
- **Expected Results:** Stock adjusted based on condition, cash deducted, return invoice generated.
- **Error Scenarios:** Return qty exceeds sale qty.
- **APIs:** `POST /api/pos/returns`

## 7. Product Exchange/Replacement Flow

- **Preconditions:** Customer returning defective unit within warranty.
- **Actor:** Manager
- **Steps:**
  1. Lookup serial/invoice.
  2. Mark old unit as `TESTING_PENDING`.
  3. Issue new unit from stock.
  4. Transfer warranty.
- **Expected Results:** Inventory swapped, warranty updated.
- **Error Scenarios:** No replacement stock available.
- **APIs:** `POST /api/inventory/exchanges`

## 8. Due Collection Flow

- **Preconditions:** Customer has `due_balance > 0`.
- **Actor:** Cashier
- **Steps:**
  1. Select Customer.
  2. Enter collection amount and payment method.
  3. Apply to specific invoices or auto-apply to oldest.
  4. Save.
- **Expected Results:** Customer due reduced, cash increased, sale status updated.
- **Error Scenarios:** Collection exceeds due.
- **APIs:** `POST /api/customers/collect-due`

## 9. EMI Creation & Payment Flow

- **Preconditions:** Customer approved for EMI.
- **Actor:** Manager
- **Steps:**
  1. At POS, select EMI payment.
  2. Set down payment, interest rate, months.
  3. Generate schedule.
  4. Accept down payment.

- **Expected Results:** EMI schedule created, initial payment recorded.
- **Error Scenarios:** Invalid math configuration.
- **APIs:** **POST** /api/pos/checkout (with EMI payload), **POST** /api/emi/pay-installment

## 10. Daily Settlement/Shift Closing Flow

- **Preconditions:** End of day/shift.
- **Actor:** Cashier/Manager
- **Steps:**
  1. Open Settlement screen.
  2. Count physical cash.
  3. Enter actual cash amount.
  4. Submit.
- **Expected Results:** Expected vs Actual calculated, discrepancy logged, shift closed.
- **Error Scenarios:** Pending unhandled transactions.
- **APIs:** **POST** /api/accounting/settlements

## 11. Expense Recording Flow

- **Preconditions:** Expense categories configured.
- **Actor:** Any staff (with permission)
- **Steps:**
  1. Select Category.
  2. Enter amount, note, and payment source.
  3. Save.
- **Expected Results:** Ledger cash reduced, Expense recorded.
- **Error Scenarios:** Insufficient funds in selected account.
- **APIs:** **POST** /api/accounting/expenses

## 12. Capital Investment & Drawing Flow

- **Preconditions:** Owner account.
- **Actor:** Owner
- **Steps:**
  1. Go to Equity.
  2. Select Invest or Draw.
  3. Enter amount and account.
- **Expected Results:** Ledger and Equity totals updated.
- **APIs:** **POST** /api/accounting/equity

## 13. Account Transfer Flow

- **Preconditions:** Multiple accounts exist (e.g., Cash, Bank).
- **Actor:** Manager/Owner

- **Steps:**
  1. Select From Account and To Account.
  2. Enter amount.
  3. Save.
- **Expected Results:** Double ledger entry created, no net P&L change.
- **Error Scenarios:** Transfer exceeds available balance.
- **APIs:** **POST** /api/accounting/transfers

## 14. Warranty Verification Flow

- **Preconditions:** None.
- **Actor:** Staff
- **Steps:**
  1. Scan/enter serial number or invoice ID.
  2. View status.
- **Expected Results:** Displays Active/Expired status and days remaining.
- **APIs:** **GET** /api/inventory/warranty-check

## 15. Service Ticket Lifecycle Flow

- **Preconditions:** Device received for repair.
- **Actor:** Technician/Manager
- **Steps:**
  1. Create ticket (RECEIVED).
  2. Update to DIAGNOSING.
  3. Add parts and labor cost (IN\_REPAIR).
  4. Mark READY and notify customer.
  5. Take payment and mark DELIVERED.
- **Expected Results:** Ticket tracks history, parts deducted upon delivery, cash collected.
- **APIs:** **POST** /api/service/tickets , **PATCH** /api/service/tickets/{id}/status

## 16. Barcode Generation & Printing Flow

- **Preconditions:** Products exist.
- **Actor:** Manager
- **Steps:**
  1. Select products and quantities.
  2. Configure label size.
  3. Print via browser print dialog.
- **Expected Results:** Labels sent to printer.
- **APIs:** Frontend specific.

## 17. Item/Serial Lookup Flow

- **Preconditions:** None.
- **Actor:** Staff
- **Steps:**
  1. Global search serial number.
  2. View product lifecycle (purchase, sale, return history).
- **Expected Results:** Complete audit trail of unit.
- **APIs:** **GET** /api/inventory/serials/{serial}

## 18. Staff User Creation & Permission Assignment Flow

- **Preconditions:** Owner/Admin access.
- **Actor:** Owner
- **Steps:**
  1. Create user (name, phone, pin).
  2. Select Role (Cashier, Manager).
  3. Customize specific permissions if needed.
- **Expected Results:** Staff can log in with scoped access.
- **APIs:** **POST** /api/users/staff

## 19. Customer Profile & History Flow

- **Preconditions:** Customer exists.
- **Actor:** Any Staff
- **Steps:**
  1. Open Customer profile.
  2. View stats, due balance, and past invoices.
- **Expected Results:** Aggregated customer data displayed.
- **APIs:** **GET** /api/customers/{id}

## 20. Report Generation & Export Flow

- **Preconditions:** Data exists.
- **Actor:** Manager/Owner
- **Steps:**
  1. Navigate to Reports.
  2. Select date range and report type (Sales, P&L, Tax).
  3. Click Export to Excel/PDF.
- **Expected Results:** File downloaded.
- **APIs:** **GET** /api/reports/sales?export=csv